COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S2	**SECTION:**	CPE21S2
DATE PERFORMED:	7/10/26	**DATE SUBMITTED:**	7/10/26
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ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

1.Compute the total and average of 3 quizzes then display the total and average.

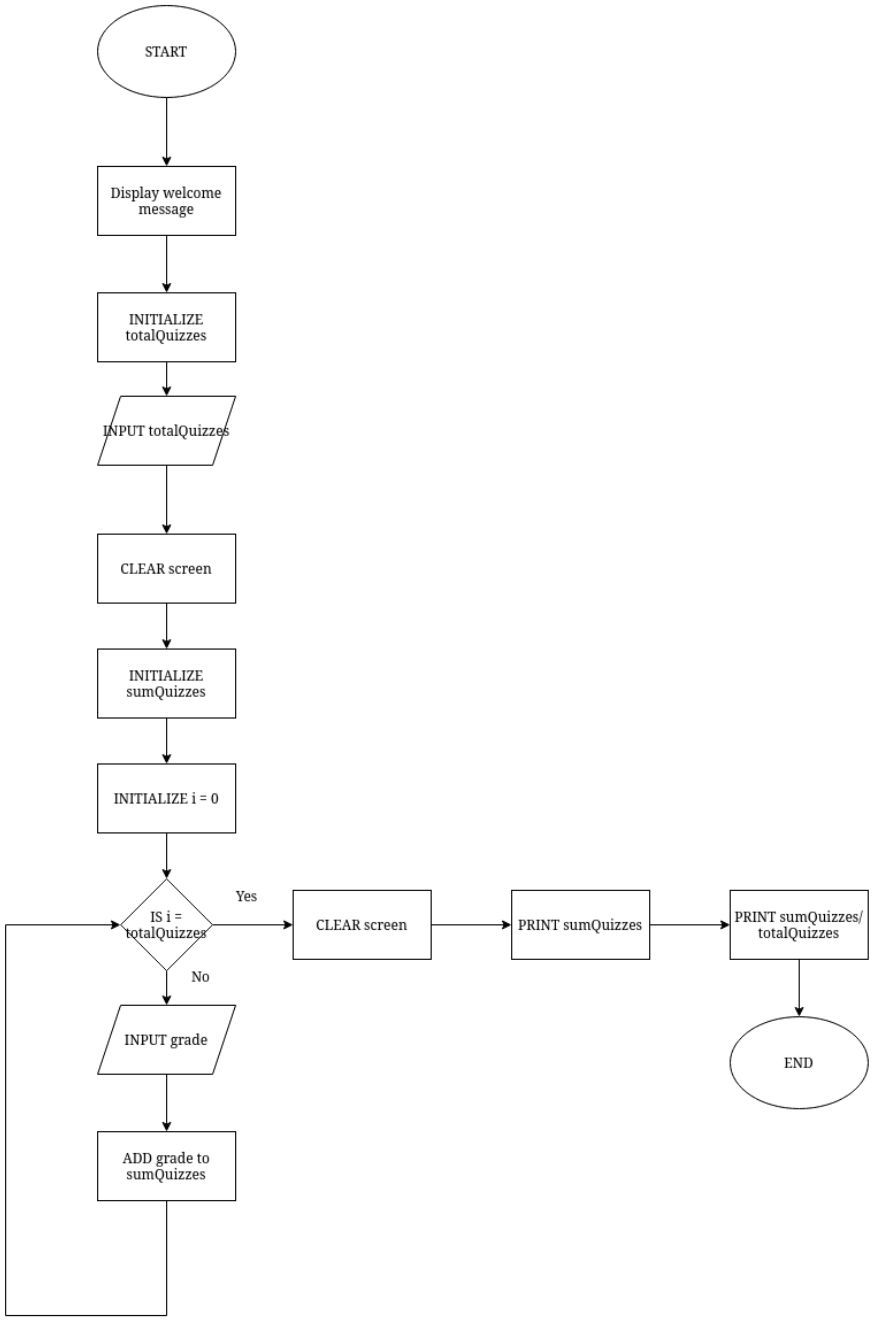
Algorithm

1.Display welcome message
2.Initialize a variable to store the sum of quizzes and to know the number of quizzes
3.Input the number of quizzes to be computed
4.Loop until the grade for all the quizzes has been inputted
5. Add all the grades together then store the sum in a variable
6. Divide the sum of the grades then divide it by the number of quizzes then display it also
7. Display the average and sum

PseudoCode

```
BEGIN
  DISPLAY welcome message
  INITIALIZE sumQuizzes,totalQuizzes
  FOR i = 0 to totalQuizzes
     PRINT a message asking the user to input grades for quizzes
     WAIT for user to INPUT the grades of the quizzes
     CREATE temporary variable to store the input
     ADD the inputted variable to sumQuizzes
  ENDFOR
  PRINT sumQuizzes with the message "Sum of quizzes is ..."
  PRINT sumQuizzes DIVIDED by totalQuizzes with the message of "Average of quizzes is ..."
END
```

Flowchart


START
Display welcome message
INITIALIZE totalQuizzes
INPUT totalQuizzes
CLEAR screen
INITIALIZE sumQuizzes
INITIALIZE i = 0
IS i = totalQuizzes
Yes
No
CLEAR screen
PRINT sumQuizzes
PRINT sumQuizzes/ totalQuizzes
END
INPUT grade
ADD grade to sumQuizzes

CODE:

```
#include <iostream>

int main(){
   std::cout << "============== Quiz Grade calculator
==============\n\n";
   float sumQuizzes = 0;
   int totalQuizzes = 0;

   std::cout << "Input number of quizzes:";
   std::cin >> totalQuizzes;

   system("clear");//Clear Screen

   std::cout << "Input the grades of the Quizzes below:\n\n";
   for (int i = 0; i < totalQuizzes; i++){
       std::cout << "Input the grades for Quiz " << i+1 << "\n";
       int grade;
       std::cin >> grade;
       sumQuizzes += grade;
   }
   system("clear");

   std::cout << "-----------------------------------------\n";
   std::cout << "Sum of Grades is " << sumQuizzes << "\n";
   std::cout << "Average of Quizzes is " << sumQuizzes/totalQuizzes <<
"\n";

   return 0;
}
```

Output:

```
~/Downloads/SW#1.1
❯ cd "/home/nielc/Downloads/SW#1.1/" && g++ main.cpp -o main && "/home/nielc/Downloads/SW#1.1/"main
============== Quiz Grade calculator ==============

Input number of quizzes:3
```

```
Input the grades of the Quizzes below

Input the grades for Quiz 1
6
Input the grades for Quiz 2
5
Input the grades for Quiz 3
9
```

```
-----------------------------------------
Sum of Grades is 20
Average of Quizzes is 6.66667
```

Code Explanation:

The program first prints a welcome message.Next, it creates a variable where it will store the sum of quizzes and the number of quizzes. It then asks the user to input the amount of quizzes they want to calculate on. The program then repeatedly asks the user about the scores of the student until it fulfills the amount of quizzes the user wants. Within this loop, the program adds each score to the variable holding the total score of quizzes. Lastly, It displays the value of the variable holding the sum of quizzes and then it also displays the average by dividing this sum by the number of quizzes.

2.Cashiering system that computes the total price of the 3 items bought, accepts payment and computes change. Display the transactions made.

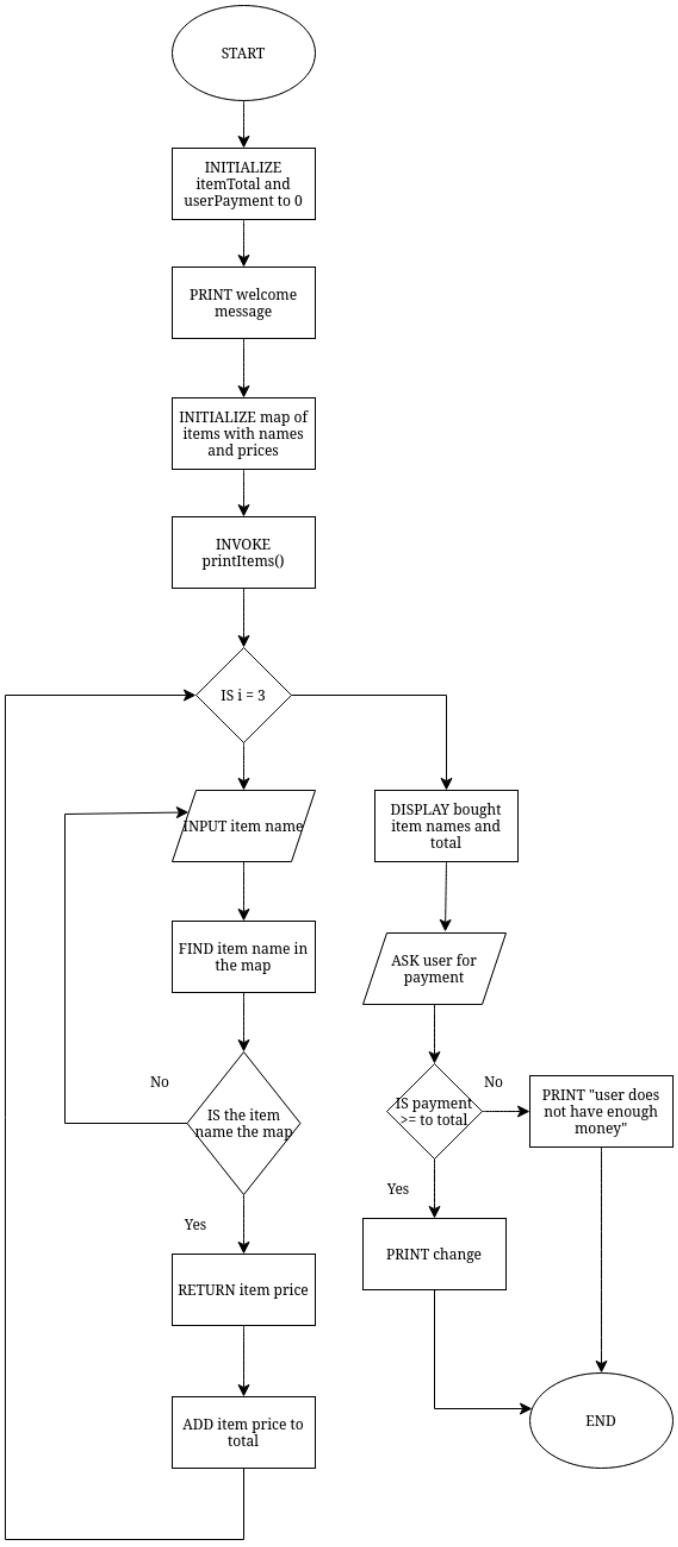
Algorithm

1. Initialize itemTotal and userPayment to 0 and itemNames array with size 3
2. Display welcome message
3. Initialize map for items
4. Print items
5. Loop until user bought 3 items
6. Compute total
7. Accept payment
8. If payment is greater than total price, compute the change, if not, print message telling the transaction is not completed
9. Print leaving message

PseudoCode

```
BEGIN
   INITIALIZE itemTotal and userPayment to 0, itemNames array
   PRINT welcome message
   INITIALIZE map of items with names and prices
   INVOKE printItems()
   FOR i = 0 to 2
      INPUT item name
      GET item price
      ADD item price to item total
      PRINT added the price to total payment
   ENDFOR
   DISPLAY itemTotal
   INPUT userPayment
   IF userPayment > or = to itemTotal
      PRINT change
   ELSE
      PRINT "User has not enough money"
END
```

Flowchart


START
INITIALIZE itemTotal and userPayment to 0
PRINT welcome message
INITIALIZE map of items with names and prices
INVOKE printItems()
IS i = 3
INPUT item name
FIND item name in the map
IS the item name the map
No
Yes
RETURN item price
ADD item price to total
DISPLAY bought item names and total
ASK user for payment
IS payment >= to total
No
PRINT "user does not have enough money"
Yes
PRINT change
END

Code:

```
#include <iostream>
#include <map>
#include <string>

void printItems(const std::map<std::string,int> &items){
   std::cout << "{ITEMS FOR SELL\n\tName:\tPrice\n";
   for (const auto& item: items){
       std::cout << "\t["<< item.first << ":\t" << item.second << "]\n";
   }
   std::cout << "}\n\n";
}

int getItemAndPrice(const std::map<std::string,int> &items,std::string
*itemNames,int currentIndex){
   std::string itemName;
   std::cout << "Enter name of item :";
   std::getline(std::cin,itemName);

   if (items.find(itemName) != items.end()){
       std::cout << "Purchased " << itemName << "\n";
       itemNames[currentIndex] = itemName;
       return items.at(itemName);
   }
   else{
       std::cout << "Item not found, please try again\n";
       return getItemAndPrice(items,itemNames,currentIndex);
   }
}

int main(){
   int itemTotal = 0;
   int userPayment = 0;
   std::string itemNames[3] = {};

   std::cout << "Kendo Gun Shop\n";
```

```
    std::map<std::string,int> items = {
        {"Glock 19",500},
        {"Smith & Wesson Shield Plus",450},
        {"Mossberg 500",500},
        {"Ruger 10/22",350},
        {"AR-15",700},
    };

    printItems(items);

    for (int i = 0; i < 3; i++){
        int itemPrice = getItemAndPrice(items,itemNames,i);
        itemTotal += itemPrice;
        std::cout << "Added $" << itemPrice << " to total\n";
    }

    std::cout << "--------------------------\nItems bought:\n";

    for (int i = 0;i < 3; i++){
        std::cout << itemNames[i] << "\t\t $" << items.at(itemNames[i])
<< "\n";
    }

    std::cout << "---------------------------------------\n";
    std::cout << "Total price is $" << itemTotal << "\n" ;
    std::cout << "Input payment:$";
    std::cin >> userPayment;

    if (userPayment >= itemTotal){
        std::cout << "User paid $" << userPayment << ". User's change is
$" << userPayment - itemTotal;
    }
    else {
        std::cout << "User does not have enough money, Items were
returned";
    }

    return 0;
```

```
}
```

Output

```
❯ cd "/home/nielc/Downloads/SW#1.1/" && g++ main_2.cpp -o main_2 && "/home/nielc/Downloads/SW#1.1/"main_2
Kendo Gun Shop
{ITEMS FOR SELL
        Name:   Price
        [AR-15: 700]
        [Glock 19:      500]
        [Mossberg 500:  500]
        [Ruger 10/22:   350]
        [Smith & Wesson Shield Plus:    450]
}

Enter name of item :AR-15
Purchased AR-15
Added $700 to total
Enter name of item :AR-15
Purchased AR-15
Added $700 to total
Enter name of item :Colt 1911
Item not found, please try again
Enter name of item :AR-15
Purchased AR-15
Added $700 to total
--------------------------
Items bought:
AR-15           $700
AR-15           $700
AR-15           $700
-------------------------------------
Total price is $2100
Input payment:$2100
User paid $2100. User's change is $0
```

```
❯ cd "/home/nielc/Downloads/SW#1.1/" && g++ main_2.cpp -o main_2 && "/home/nielc/Downloads/SW#1.1/"main_2
Kendo Gun Shop
{ITEMS FOR SELL
        Name:   Price
        [AR-15: 700]
        [Glock 19:      500]
        [Mossberg 500:  500]
        [Ruger 10/22:   350]
        [Smith & Wesson Shield Plus:    450]
}

Enter name of item :Glock 19
Purchased Glock 19
Added $500 to total
Enter name of item :Glock 19
Purchased Glock 19
Added $500 to total
Enter name of item :Glock 19
Purchased Glock 19
Added $500 to total
--------------------------
Items bought:
Glock 19                $500
Glock 19                $500
Glock 19                $500
-------------------------------------
Total price is $1500
Input payment:$100
User does not have enough money, Items were returned

~/Downloads/SW#1.1 9s
```

Code Explanation:

The program first prints the name of the shop and the items, its name and price. This program uses a map to store the name as a string and price as an integer in one location.After that the program loops 3 times asking the user what they want to buy. This prompt is case sensitive- capitalizations matter. If the product is not on the list, The program asks the user again until the product is found. This does not affect the number of overall items. Next, the program displays the bought items' names, price , and the total price. Lastly,it asks the user for payment. If the payment of the user is greater than or equal to the total price, the program lets the user purchase the items, it also displays the change. If not, it tells the user that the purchase is not successful and the items were returned.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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